

CustomerIQ

Product Requirements Document & Official Project Record

Version 4.0 | Project Status: End of Day 5

1. Purpose

Official consolidated CustomerIQ record through Day 5. It combines the previous PRD through Day 4 with the complete Day 5 predictive-modeling milestone, preserving scope, methodology, datasets, results, artifacts, assumptions, risks, limitations and next steps.

2. Executive Summary

CustomerIQ is an AI-powered customer intelligence and retention analytics system built from transaction-level retail data. The project follows: Customer Behavior Insight → Risk Identification → Priority → Action → Monitoring. The goal is a business-first, reproducible system rather than a model built in isolation.

- Day 1: data foundation, validation, cleaning, revenue and baseline KPIs.
- Day 2: customer order behavior, purchase intervals, frequency tiers, right-censoring and temporal cutoff.
- Day 3: repeat-customer cohort, pre-cutoff features, 90-day return/no-return outcome and leakage audit.
- Day 4: retention-driver comparisons, validation, skew/outlier diagnostics and visual analysis.
- Day 5: leakage-safe Logistic Regression baseline, held-out predictions, probabilities and model evaluation.

3. Product Vision, Problem and Users

Area	Definition
Working description	AI-Powered Customer Intelligence & Retention Analytics Engine
Primary problem	Turn transaction history into actionable retention priorities.
Primary users	Marketing analysts, CRM/retention teams, analytics teams and retail/e-commerce managers
Secondary users	Account managers, business analysts and executives
Core jobs	Understand behavior; identify disengagement; estimate observed churn risk; prioritize resources; explain risk; monitor customer health
Demonstration context	UCI Online Retail II with meaningful wholesale/B2B characteristics

4. Dataset Evidence Base

Metric	Observed result
Dataset	UCI Online Retail II
Raw rows / columns	1,067,371 / 8
Date range	2009-12-01 07:45:00 to 2011-12-09 12:50:00
Unique customers before cleaning	5,942
Unique invoices before cleaning	53,628

Metric	Observed result
Fields	Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country
Missing Description	4,382
Missing Customer ID	243,007 (22.77%)
Exact duplicates	34,335

Interpretation rule: wholesale/B2B purchasing behavior means results should not be generalized as universal consumer e-commerce benchmarks.

5. Day 1 — Data Foundation and Cleaning — COMPLETE

- Validated shape, columns, types, missing values, duplicates, dates, customers and invoices.
- Removed exact duplicates, missing Customer IDs, cancelled invoices beginning with C, non-positive quantities and non-positive prices.
- Converted InvoiceDate to datetime and derived Revenue = Quantity × Price for analysis.
- Saved online_retail_cleaned.csv and calculated baseline KPIs.

Metric	Observed result
Total Revenue	£17,374,804.27
Total Customers	5,878
Total Orders	36,969
Average Order Value	£469.98
Repeat Customers	4,255
One-Time Customers	1,623
Repeat Rate	72.39%
Cleaned dataset	779,425 rows × 8 original fields

6. Day 2 — Purchase Behavior and Temporal Methodology — COMPLETE

Customer order distribution: minimum 1, median 3.0, average 6.29 and maximum 398 orders. Inter-purchase intervals were analyzed; provisional tiers were 2–3, 4–9 and 10+, with one-time customers tracked separately. Right-censoring near the dataset end was explicitly recognized.

Temporal design	Decision
Observation cutoff	2011-09-10
Validation window	2011-09-10 through 2011-12-09
Look-forward period	90 days
Dataset end	2011-12-09

The 90-day window is an observation/validation window, not a universal definition of permanent churn.

7. Day 3 — Churn Methodology and Customer-Level Dataset — COMPLETE

Metric	Observed result
Customers before cutoff	5,281
One-time pre-cutoff	1,576 (29.84%)

Metric	Observed result
Eligible 2+ order customers	3,705 (70.16%)
Returned during validation	1,973 (53.25%)
No observed return	1,732 (46.75%)
Provisional observed churn	46.75%
Final dataset	3,705 rows × 11 columns

Fields: Customer ID, Frequency, Monetary, First Purchase Date, Last Purchase Date, Customer Lifespan, Recency, Average Purchase Interval, Median Purchase Interval, returned_in_validation and observed_churn.

Customer summary	Observed result
Average Frequency	7.76
Median Frequency	4
Average Monetary	£3,615.15
Median Monetary	£1,333.70
Average Lifespan	319.5 days
Average Recency	155.5 days

Leakage audit: PASS. All predictive features use only pre-cutoff transactions; the outcome uses only the 90-day validation period. No future purchases were used in feature construction. One row per eligible customer was validated. Output: Data/Processed/customer_churn_analysis.csv.

8. Day 4 — Retention Driver Analysis — COMPLETE

Check	Result
Rows / columns	3,705 / 11
Returned	1,973
No observed return	1,732
Missing	0
Duplicate IDs	0
Infinite values	0
Invalid negative feature values	0
Fresh-kernel execution	PASS

Feature	Median difference for returned customers
Monetary	+£1,182.92
Customer Lifespan	+154 days
Recency	−139 days
Average Purchase Interval	−25.10 days
Median Purchase Interval	−22 days
Frequency	+3 orders

- Monetary and Frequency were especially skewed.

- Skewness and outliers were reported without automatic removal.
- Clearest descriptive signals: Monetary, Customer Lifespan and Recency.
- Results show association, not causation.
- Day 4 served as a plausibility check before predictive modeling.

9. Day 5 — Predictive Modeling Baseline — COMPLETE

Day 5 established CustomerIQ's first predictive model. Logistic Regression was selected as an interpretable, reproducible baseline before testing more complex models.

Input Contract

Item	Specification
Input file	Data/Processed/customer_churn_analysis.csv
Input shape	3,705 customers × 11 columns
Customer grain	One row per eligible repeat customer
Target	observed_churn
Target meaning	1 = no observed return; 0 = observed return
Positive class	Observed churn = 1
Missing feature values	0
Duplicate Customer IDs	0

Six Predictors Used

Feature	Meaning
Frequency	Unique historical orders before cutoff
Monetary	Historical customer monetary value before cutoff
Recency	Time since last purchase before cutoff
Customer Lifespan	Duration between first and last purchase
Average Purchase Interval	Average days between purchases
Median Purchase Interval	Typical purchase gap, less sensitive to extremes

Leakage Prevention

- X contains only the six pre-cutoff behavioral predictors.
- returned_in_validation was excluded because it comes directly from the validation window.
- observed_churn was used only as the target.
- Customer ID was excluded as an identifier rather than behavior.
- No future transaction or validation-period information was used as a predictor.
- StandardScaler was placed inside the training pipeline so preprocessing is learned from training data only.

Model Workflow

Customer-level dataset → predictor selection → target selection → stratified train/test split → StandardScaler inside pipeline → Logistic Regression → held-out predictions and probabilities → evaluation → coefficient interpretation → prediction export.

Component	Decision
Model	Logistic Regression
Preprocessing	StandardScaler inside model pipeline
Split	Stratified train/test split

Component	Decision
Training customers	2,964
Held-out test customers	741
Target distribution	{0: 1973, 1: 1732}
Metrics	Accuracy, Precision, Recall, F1, ROC-AUC
Output	Predicted class and churn probability

Actual Day 5 Results

Metric	Result
Accuracy	0.7045 (70.45%)
Precision	0.7068 (70.68%)
Recall	0.6272 (62.72%)
F1 Score	0.6646 (66.46%)
ROC-AUC	0.7731 (77.31%)

The baseline shows useful discrimination, but it is not production-ready. The business cost of false positives versus false negatives and the operating decision threshold have not yet been optimized.

Coefficient Findings

Finding	Observed result
Strongest positive churn coefficient	Average Purchase Interval: +0.8286
Strongest negative churn coefficient	Median Purchase Interval: -0.4052

Coefficients are predictive associations on standardized features, not causal effects. The two interval variables describe related purchase-rhythm information and may be correlated, so their directions should not be interpreted as isolated causal rules.

Prediction Output and Deliverables

Artifact	Purpose / status
Notebook/05_predictive_modeling.ipynb	Complete Day 5 workflow; successfully executed from a fresh kernel
Data/Processed/logistic_regression_predictions.csv	Held-out prediction export
Prediction rows	741 test customers
Prediction content	Actual outcome, predicted outcome and churn probability
Source datasets	customer_churn_analysis.csv and online_retail_cleaned.csv remained unmodified

Validation and Reproducibility

- Required packages were available in the project virtual environment, including pandas, matplotlib, scikit-learn, ipykernel and nbconvert.
- Fresh-kernel end-to-end execution succeeded.
- Prediction CSV was created successfully with 741 held-out rows.

- Schema, row-count and split checks guarded against silent workflow errors.
- Day 4 data-quality checks confirmed complete numeric predictive features.
- Source CSVs were not modified.

10. Current Project Architecture

Artifact	Role / status
Day 1 notebook	Validation, cleaning, revenue and KPIs — COMPLETE
02_purchase_behavior.ipynb	Orders, intervals, tiers and cutoff — COMPLETE
03_churn_methodology.ipynb	Cohort, features, outcome and leakage checks — COMPLETE
04_retention_analysis.ipynb	Retention comparisons and diagnostics — COMPLETE
05_predictive_modeling.ipynb	Logistic Regression baseline and prediction probabilities — COMPLETE
online_retail_cleaned.csv	Reusable cleaned transaction source
customer_churn_analysis.csv	Customer-level analytical dataset
logistic_regression_predictions.csv	Held-out Day 5 prediction output

11. Progress Assessment

Days 1–5 are complete. CustomerIQ now has a data foundation, time-aware churn methodology, customer-level dataset, descriptive retention analysis and its first interpretable predictive baseline. Planning estimate: approximately 45–50% of the full project.

12. Next: Day 6

Random Forest benchmark and controlled comparison with Logistic Regression using the same cohort, target definition and predictor contract.

- Compare Accuracy, Precision, Recall, F1, ROC-AUC and confusion matrix.
- Inspect feature importance without confusing predictive importance with causation.
- Choose the model using performance, interpretability, robustness and business usefulness.

13. Remaining Roadmap to 15–17 Days

Day	Deliverable
6	Random Forest benchmark and model comparison
7	RFM scoring and interpretable segmentation
8	Customer Value × Predicted Risk prioritization
9	Traceable recommendation / decision rules
10	Selected SQL analyses
11–12	Executive and priority/action dashboard in Power BI
13	GitHub repository, README and reproducibility packaging
14	Portfolio case study, screenshots and narrative
15	Lightweight MVP/prototype if analytics are stable
16	End-to-end testing and polish

Day	Deliverable
17	Final PRD, demo, presentation and interview package

14. Risks and Limitations

- Right-censoring remains because the dataset ends on 2011-12-09.
- No observed return in 90 days is a provisional study outcome, not universal permanent churn.
- Wholesale/B2B behavior limits generalization.
- Low-order customers provide noisier interval evidence.
- Descriptive associations and model coefficients do not prove causation.
- Monetary and Frequency are skewed.
- Logistic Regression is only a baseline; no model comparison has yet been completed.
- Decision thresholds have not been optimized for business cost or intervention capacity.
- Predictions indicate statistical risk patterns, not certainty.
- Recommendations remain hypotheses/rules unless causal evidence is established.

15. Product Success Criteria

- Reproducible pipeline and documented cleaning.
- Defensible time-aware methodology and leakage control.
- Evaluation beyond simple accuracy.
- Model comparison before final model selection.
- Interpretable segmentation.
- Customer value combined with risk.
- Traceable recommendation logic.
- Focused Power BI dashboard.
- Clear GitHub README and portfolio narrative.
- Ability to explain decisions, metrics and limitations in an interview.

16. Updated Project Narrative

I built CustomerIQ, a customer intelligence and retention analytics system using transaction-level UCI Online Retail II data. I validated and cleaned more than one million raw records, created a reusable transaction dataset, analyzed order frequency and inter-purchase intervals, and used a fixed observation cutoff plus a 90-day look-forward period to address the dataset end-date problem. I transformed pre-cutoff history into a customer-level dataset for 3,705 eligible repeat customers and created a time-separated observed return/no-return outcome with an explicit leakage audit. I compared returned and non-returned customers across Frequency, Monetary value, Recency, Customer Lifespan and purchase intervals, then built a leakage-safe Logistic Regression baseline using six behavioral features. On a held-out test set of 741 customers, the model achieved 70.45% accuracy, 70.68% precision, 62.72% recall, 66.46% F1 and ROC-AUC of 0.7731 while producing customer-level churn probabilities. Next comes Random Forest comparison, segmentation, value-risk prioritization, recommendation rules and a business-facing prototype/dashboard.

17. Official Change Log

Version	Project state	Change
1.0	End of Day 1	Initial PRD
2.0	End of Day 3	Added Day 2 methodology and Day 3 cohort, features, outcome and leakage audit

Version	Project state	Change
3.0	End of Day 4	Added retention analysis, exact comparisons, validation and roadmap
4.0	End of Day 5	Added complete Logistic Regression baseline, predictor contract, leakage exclusions, exact metrics, coefficient findings, 741 held-out predictions, reproducibility checks and Day 6 handover

End of official project record — CustomerIQ PRD v4.0.